Complex Numbers

ANSWER KEY

Section 1: Cartesian Form & Complex Arithmetic

1 Given $z = 3 - 4i$, compute $z \cdot z^*$ where z^* is the complex conjugate of z.

a. **25** ✓

b. 7

c. -7

d. $25 - 24i$

$z^* = 3 + 4i$. Then $z \cdot z^* = 3^2 - (4i)^2 = 9 - (-16) = 25$.

2 Express $\dfrac{5 + 2i}{2 - i}$ in the form $a + bi$.

a. $\frac{8}{5} + \frac{9}{5}i$ ✓

b. $\dfrac{12}{5} + \dfrac{9}{5}i$

c. $8 + 9i$

d. $\dfrac{8}{3} + \dfrac{9}{3}i$

Multiply numerator and denominator by $(2 + i)$: $\dfrac{(5 + 2i)(2 + i)}{4 + 1} = \dfrac{10 + 5i + 4i + 2i^2}{5} = \dfrac{8 + 9i}{5} = \frac{8}{5} + \frac{9}{5}i$.

3 Find the two square roots of $-5 + 12i$.

a. $\pm(2 + 3i)$ ✓

b. $\pm(3 + 2i)$

c. $\pm(2 - 3i)$

d. $\pm(3 - 2i)$

Let $(x + iy)^2 = -5 + 12i \Rightarrow x^2 - y^2 = -5$ and $2xy = 12 \Rightarrow xy = 6$. Also $x^2 + y^2 = \sqrt{25 + 144} = 13$. Adding: $2x^2 = 8 \Rightarrow x = \pm 2$. Then $y = 6/x = \pm 3$. Roots: $\pm(2 + 3i)$.

Section 2: Modulus-Argument Form

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4 Find the modulus and principal argument of $z = -1 + i\sqrt{3}$.

a. $r = 2, \theta = \dfrac{2\pi}{3}$ ✓

b. $r = 2, \theta = \dfrac{\pi}{3}$

c. $r = 4, \theta = \dfrac{2\pi}{3}$

d. $r = 2, \theta = -\dfrac{2\pi}{3}$

Modulus $r = \sqrt{(-1)^2 + (\sqrt{3})^2} = \sqrt{4} = 2$. Point $(-1, \sqrt{3})$ is in QII: $\theta = \pi - \arctan(\sqrt{3}) = \pi - \pi/3 = 2\pi/3$.

5 Given $z_1 = 3(\cos(\pi/4) + i\sin(\pi/4))$ and $z_2 = 2(\cos(\pi/6) + i\sin(\pi/6))$, find $\arg(z_1 z_2)$.

a. $\dfrac{5\pi}{12}$ ✓

b. $\dfrac{\pi}{12}$

c. $\dfrac{\pi}{24}$

d. $\dfrac{7\pi}{12}$

When multiplying complex numbers, arguments add: $\arg(z_1 z_2) = \arg(z_1) + \arg(z_2) = \frac{\pi}{4} + \frac{\pi}{6} = \frac{5\pi}{12}$.

6 Express $z = 4e^{i\pi/3}$ in Cartesian form $x + iy$.

a. $2 + 2i\sqrt{3}$ ✓

b. $2\sqrt{3} + 2i$

c. $2 - 2i\sqrt{3}$

d. $4 + 4i\sqrt{3}$

$4(\cos(\pi/3) + i\sin(\pi/3)) = 4(1/2 + i\sqrt{3}/2) = 2 + 2i\sqrt{3}$.

Section 3: Loci in the Argand Plane

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7 Describe the locus of points represented by $|z - 3 - 4i| = 5$ in the complex plane.

a. **A circle with center $(3, 4)$ and radius 5 ✓**

b. A circle with center $(-3, -4)$ and radius 5

c. A circle with center $(3, 4)$ and radius 25

d. A straight line perpendicular to $3 + 4i$

$|z - (3 + 4i)| = 5$ represents all points at distance 5 from $3 + 4i$, which is a circle centered at $(3, 4)$ with radius 5.

8 Describe the locus of points satisfying $|z - 2| = |z - 6i|$.

a. **Perpendicular bisector of the segment joining $(2, 0)$ and $(0, 6)$ ✓**

b. Circle passing through $(2, 0)$ and $(0, 6)$

c. Ellipse with foci at $(2, 0)$ and $(0, 6)$

d. Line segment joining $(2, 0)$ and $(0, 6)$

The equation $|z - z_1| = |z - z_2|$ represents the perpendicular bisector of the line segment joining z_1 and z_2.

9 Describe the geometric locus defined by $\arg(z - 2) = \frac{\pi}{4}$.

a. **A half-line starting from $(2, 0)$ making an angle of $\pi/4$ with the positive real axis ✓**

b. A full line through $(2, 0)$ with gradient 1

c. A circle of radius 2

d. A ray towards the origin

$\arg(z - z_0) = \alpha$ represents a ray (half-line) emanating from z_0 at angle α, excluding the endpoint z_0.

Section 4: Roots of Polynomials & Conjugate Pairs

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10 Given that $z = 1 + 2i$ is a root of $z^3 - 3z^2 + 7z - 5 = 0$, what are the other two roots?

a. $z = 1 - 2i$ **and** $z = 1$ ✓

b. $z = -1 - 2i$ and $z = 1$

c. $z = 1 - 2i$ and $z = 5$

d. $z = -1 + 2i$ and $z = -1$

Since coefficients are real, the conjugate $1 - 2i$ must also be a root. Product of roots: $(1 + 2i)(1 - 2i) = 5$. Since constant term is -5, the third root is $5/5 = 1$.

11 Find a quadratic equation with real coefficients having root $z = 4 - 3i$.

a. $z^2 - 8z + 25 = 0$ ✓

b. $z^2 + 8z + 25 = 0$

c. $z^2 - 8z + 7 = 0$

d. $z^2 - 6z + 25 = 0$

Roots are $4 \pm 3i$. Sum $= 8$. Product $= 4^2 + 3^2 = 25$. Equation is $z^2 - 8z + 25 = 0$.

12 Find the least value of $|z|$ for points z satisfying $|z - 5 - 12i| \leq 3$.

a. **10** ✓

b. 13

c. 16

d. 7

Center of circle is $5 + 12i$, distance from origin is $\sqrt{25 + 144} = 13$. The minimum distance from origin to circle is $13 - r = 13 - 3 = 10$.

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